Estimating resting-state and task-based functional connectivity in autism spectrum disorder

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1. Covariance regression for task connectivity analysis
   - Review: the general linear model of fMRI activation
   - General linear model of covariance regression and extensions
   - Novel application: Covariance regression for fMRI
   - Application: movie-watching task connectivity

2. Selection bias and motion quality control in resting-state connectivity
   - Motion quality control creates selection bias
   - Nonparametric motion control in functional connectivity studies in children with autism spectrum disorder
   - MoCo applied to ADHD differences in the ABCD Dataset

Which brain regions are involved in naturalistic movie watching?

Roadmap for this part:

1. Classic approach: general linear model (GLM) → effect of the task on the *mean* BOLD signal.
2. Covariance regression → effect of the task on the *covariance* between brain regions.

Motivating scientific question:

- Naturalistic movie watching is child-friendly: engaging, better compliance, and sometimes less head motion than long controlled tasks or rest (Vanderwal et al. 2019).
- **This talk:** which parts of the brain are involved in naturalistic movie watching?
- **Future work:** how does autism modify the regions involved?

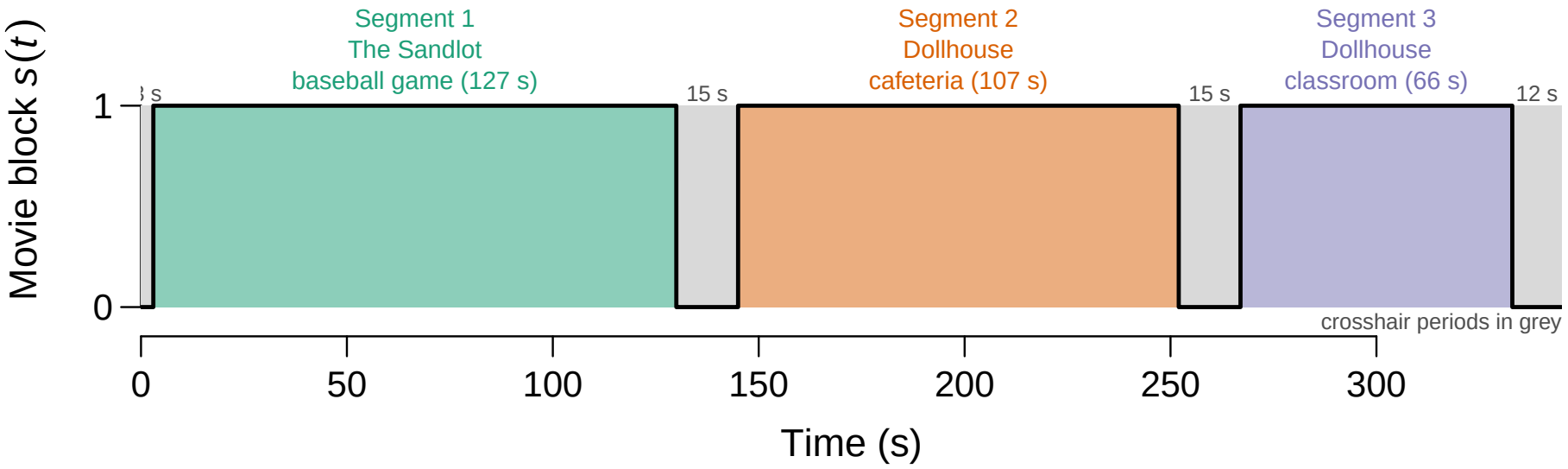
Naturalistic movie-watching task

- Children view movie excerpts on a screen through a mirror mounted on the head coil.
- Task 1 (345 s), three segments depicting social interactions among school-age children:
  - **Segment 1** (127 s, *The Sandlot*, 1993): baseball game — teamwork, failure, and shame.
  - **Segment 2** (107 s, *Welcome to the Dollhouse*, 1995): cafeteria — a girl's social isolation during lunch.
  - **Segment 3** (66 s, *Welcome to the Dollhouse*): classroom test — peer conflict and moral judgment.





Task 1 block design



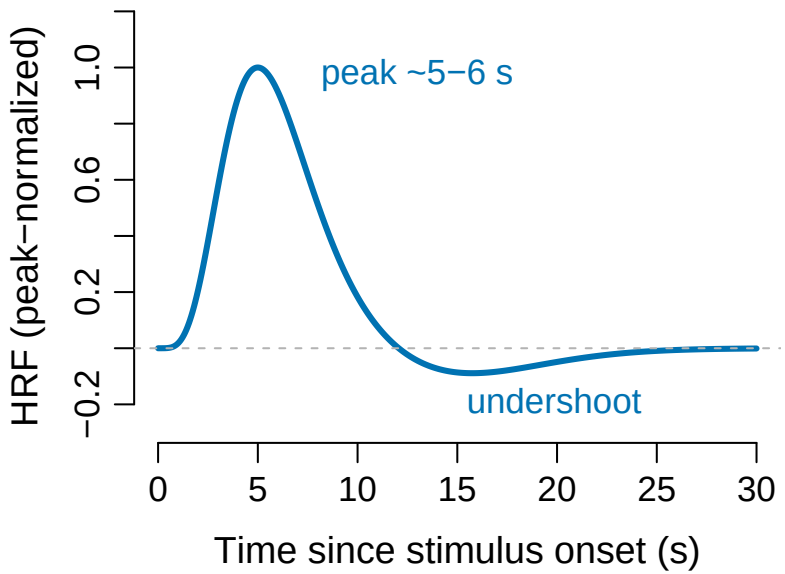
- Crosshair (fixation cross) periods: 3 s at the start, 15 s between segments, 12 s at the end.
- Movie-block indicator: $s(t) = 1$ during a movie segment, 0 otherwise.

Classic approach: creating the predictor from the task design

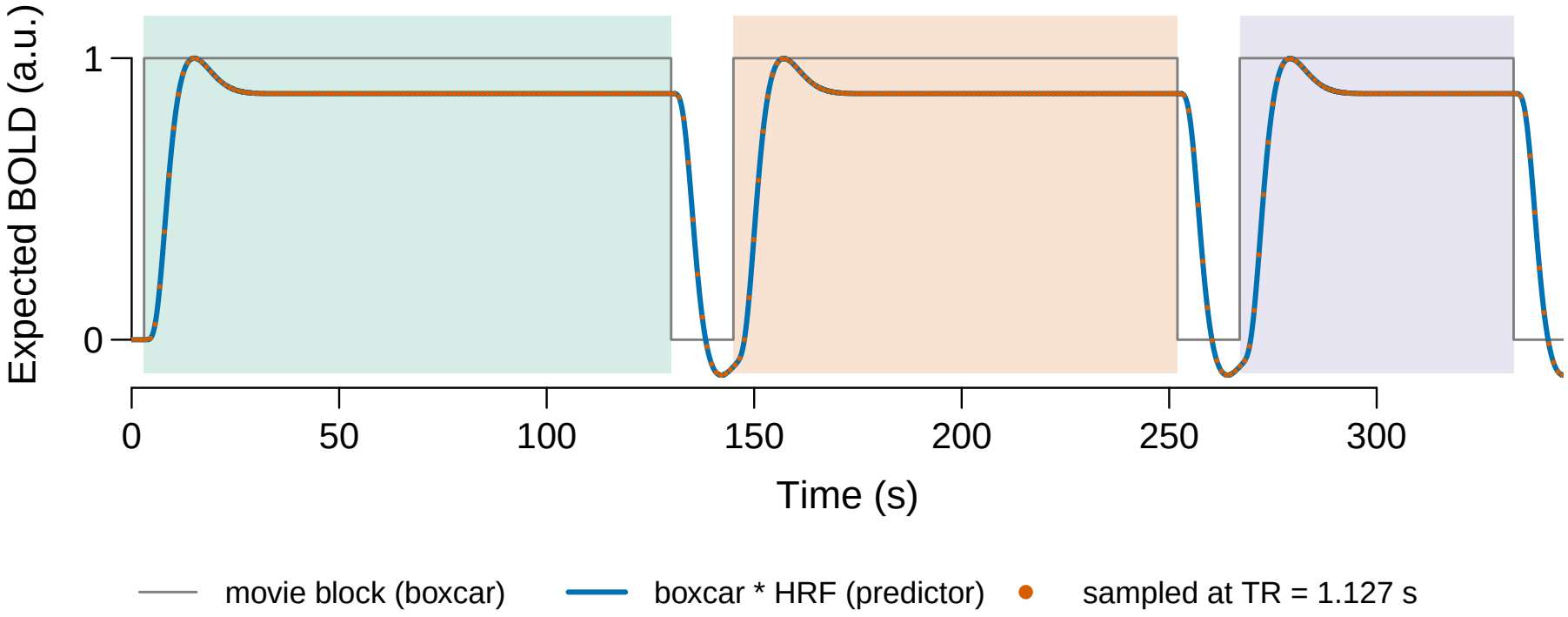
- 5-6 seconds after neural activity, the blood-oxygen-level-dependent (BOLD) response peaks.
- Stimulus function $s(t)$: task onset and duration (boxcar).
- Canonical double-gamma hemodynamic response function (HRF) $h(t)$ (Ashburner et al. 2021).
- Predictor of interest: convolution

$$x(t) = (s * h)(t) = \int_0^t s(t-u)\, h(u)\, du,$$

sampled at the acquisition times ($TR = 1.127$ s).



Predictor of interest: movie blocks ∗ HRF



- The convolved predictor lags the block onsets by ~5–6 s and dips below baseline (undershoot) after each block ends.
- Each acquisition time gives one row of the design matrix $\boldsymbol{X}$.

First-level model with AR errors

For each participant $i = 1, \ldots, n$ and each vertex/voxel v, with T volumes:

$$\boldsymbol{y}_{iv} = \boldsymbol{X}\beta_{iv} + \boldsymbol{Z}_i\boldsymbol{\delta}_{iv} + \boldsymbol{\epsilon}_{iv}, \qquad \boldsymbol{\epsilon}_{iv} \sim N(\boldsymbol{0}, \sigma^2_{iv}\boldsymbol{V}(\phi_{iv})),$$

- $\boldsymbol{X} \in \mathbb{R}^{T\times 1}$: movie-block predictor (boxcar $*$ HRF), same design for all participants; β_{iv}: activation amplitude (% signal change).
- $\boldsymbol{Z}_i \in \mathbb{R}^{T\times 15}$: intercept; head motion (translation x, y, z; rotation x, y, z) and their quadratics; linear and quadratic time.
- AR(1): $\epsilon_{it,v} = \phi_{iv}\epsilon_{i,t-1,v} + w_{it,v}$, $w_{it,v} \sim N(0, \sigma^2_{w,iv})$.
- $\hat{\phi}_{iv}$ from the residual lag-1 autocorrelation with Worsley's reduced-bias estimator (Worsley et al. 2002).
- Prewhitening (generalized least squares): $\boldsymbol{X}^*_{iv} = \hat{\boldsymbol{V}}^{-1/2}_{iv}\boldsymbol{X}$, $\boldsymbol{y}^*_{iv} = \hat{\boldsymbol{V}}^{-1/2}_{iv}\boldsymbol{y}_{iv}$; then $\hat{\beta}_{iv} = (\boldsymbol{X}^{*\top}_{iv}\boldsymbol{X}^*_{iv})^{-1}\boldsymbol{X}^{*\top}_{iv}\boldsymbol{y}^*_{iv}$.

Keep in mind: the residuals $\hat{\boldsymbol{\epsilon}}_{iv}$ from this first-level model are the input to covariance regression.

Group inference

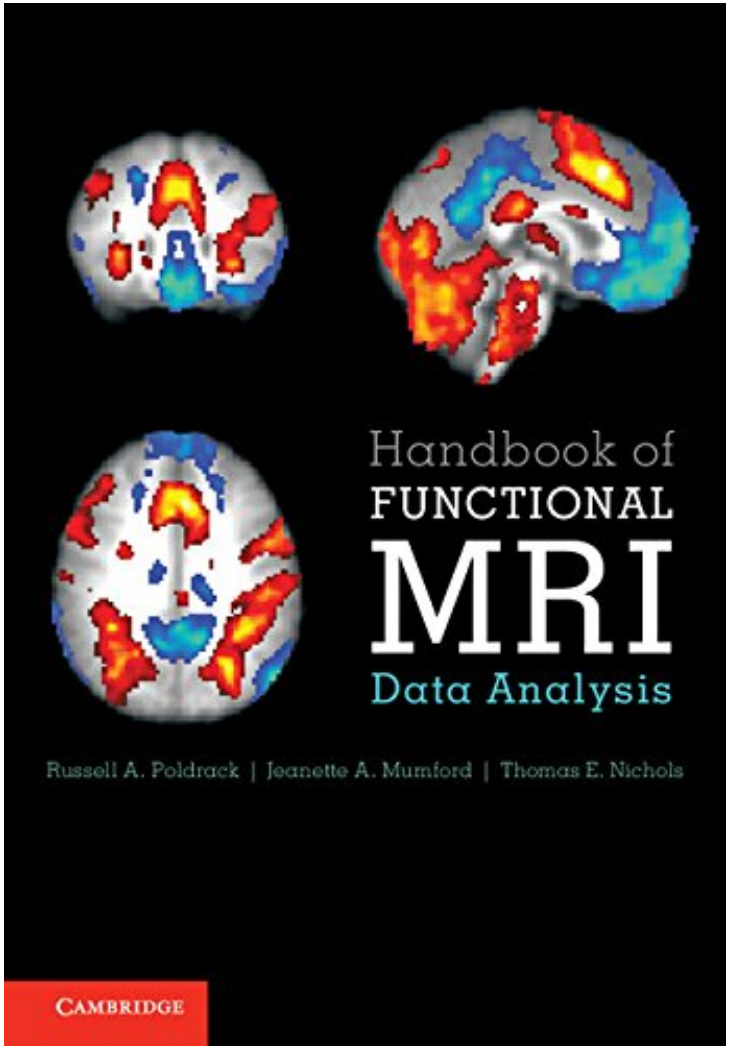
- First level yields a contrast estimate $\hat{\beta}_{iv}$ for each participant i and location v.
- Second level (massive univariate): for each v,

$$\hat{\beta}_{iv} = \beta_v + b_{iv}, \qquad H_0 : \beta_v = 0 \text{ (one-sample } t\text{-test across participants).}$$

- Multiple comparisons across thousands of locations, control FWER using random field theory, cluster-wise thresholding, or permutation tests.
- Alternative briefly mentioned here: spatial Bayesian GLM on the cortical surface (`BayesfMRI`), which pools information across neighboring vertices; joint posterior probability excursion sets control for multiple comparisons (Mejia et al. 2020).

Core reading

- The *Handbook of Functional MRI Data Analysis* (Poldrack et al. 2011) is a good introduction to this material: the HRF, first-level GLMs, group inference, and multiple comparisons.



Participants and data acquisition

- This talk: non-ASD children, 8–13 years old, Atlanta metropolitan area.
- Analytic sample: $n = 36$.
- Mock scanner training with real-time head-motion feedback (MoTrak) before scanning.
- 3T Siemens PrismaFit (FERN, Emory); fMRI: 2.4 mm isotropic, TR = 1.127 s, multiband factor 4; preprocessing: fMRIPrep.

		$n = 36$
Age, mean (SD)		11.5 (1.5)
Sex	Female	13 (36%)
	Male	23 (64%)
Race	Asian	2 (6%)
	Black	5 (14%)
	>1 race	8 (22%)
	White	21 (58%)

Group activation: movie-block predictor

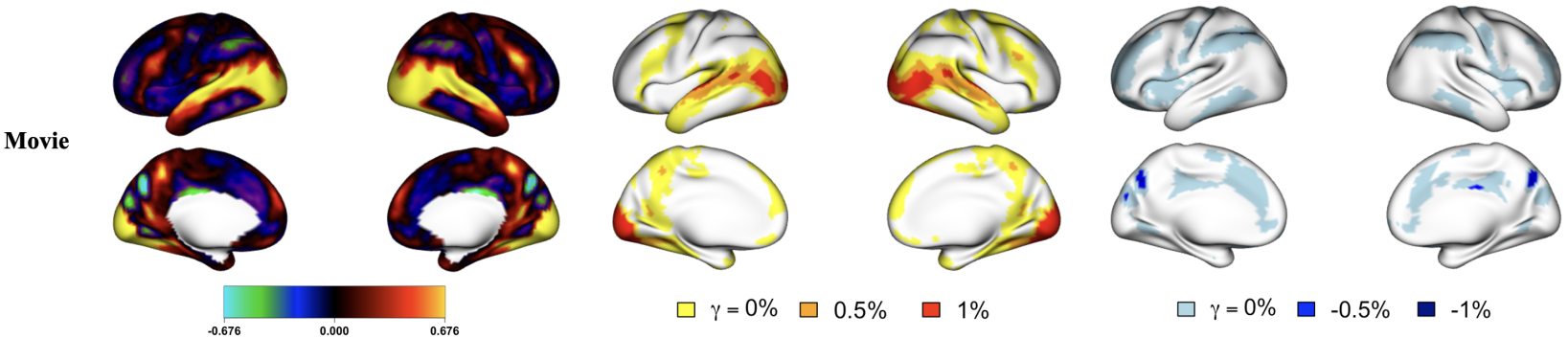


Figure: Results fit using the BayesfMRI R package (Mejia et al. 2020, Spencer et al. 2022); visualizations with ciftiTools (Pham et al. 2022).

- Activation (γ = amplitude threshold, % signal change): movie blocks versus crosshair engage visual, auditory, and ventral attention regions, including the temporal-parietal junction.

From mean activation to connectivity

- The GLM answers: how does the task change the *mean* of the BOLD signal at each location?
- The residuals $\hat{\boldsymbol{\epsilon}}_t \in \mathbb{R}^V$ (V regions) may contain additional information: the *covariance* between regions.
- Our scientific question: how do brain *connections* vary during movie watching?

Next: covariance regression — modeling the effect of task predictors on the time-varying covariance.

Covariance regression for analyzing task fMRI connectivity

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The matrix-logarithmic covariance model

Independent subjects $i = 1, \ldots, n$ with multivariate response $\boldsymbol{Y}_i = (Y_{i1}, \ldots, Y_{iV})^\top \in \mathbb{R}^V$. Mean model and a general linear model on the **log covariance matrix** (Chiu et al. 1996):

$$\boldsymbol{Y}_i \sim N_V\Big(\sum_{j=1}^{J} x_{ij}\, \boldsymbol{\beta}_j + \sum_{k=1}^{K} z_{ik}\, \boldsymbol{\gamma}_k,\ \ \boldsymbol{\Sigma}_i \Big), \qquad \log \boldsymbol{\Sigma}_i = \sum_{j=1}^{J} x_{ij}\, \boldsymbol{\alpha}_j .$$

- x_{ij}: covariates of interest ($x_{i1} = 1$ intercept); z_{ik}: nuisance covariates.
- Mean effects $\boldsymbol{\beta}_j, \boldsymbol{\gamma}_k \in \mathbb{R}^V$; covariance effects $\boldsymbol{\alpha}_j \in \mathbb{R}^{V \times V}$, symmetric: the effect of covariate j on each entry of the log covariance matrix.
- Matrix logarithm via the eigendecomposition $\boldsymbol{\Sigma}_i = \boldsymbol{U}_i \boldsymbol{\Lambda}_i \boldsymbol{U}_i^\top$: $\log \boldsymbol{\Sigma}_i = \boldsymbol{U}_i \log(\boldsymbol{\Lambda}_i)\, \boldsymbol{U}_i^\top$.
- Key property: $\boldsymbol{\Sigma}_i = \exp\big(\sum_j x_{ij}\, \boldsymbol{\alpha}_j\big)$ is positive definite for *any* symmetric $\boldsymbol{\alpha}_j \Rightarrow$ the $\boldsymbol{\alpha}_j$ are unconstrained. ($V = 1$: log-linear variance model.)
- Chiu et al. (1996) (JASA): maximum likelihood, **independent observations** — not time series, not fMRI.

Covariance regression: related approaches

- **Latent factors with quadratic covariate effects** (Hoff and Niu 2012): $\boldsymbol{\Sigma}_i = \boldsymbol{\Psi} + \boldsymbol{B}\boldsymbol{x}_i\boldsymbol{x}_i^\top\boldsymbol{B}^\top$, induced by a random-effects model; Bayesian nonparametric extension with Gaussian-process latent factors (Fox and Dunson 2015).
- **Covariate-assisted principal regression** (Zhao et al. 2021b): under a common eigenstructure, covariance matrices share eigenvectors while log eigenvalues are modeled as functions of predictors; high-dimensional extension in Zhao et al. (2021a).
- **High-dimensional sparse covariance regression** (Kim and Zhang 2026): constrained estimation on the symmetric positive-definite matrices with sparsity for high-dimensional V.
- **Bayesian dynamic covariance and correlation models** (Lan et al. 2020, Li et al. 2019): time-varying correlations via latent Gaussian processes — no predictors.
- **Other directions**: covariance is a scaled sum of matrix predictors (Zou et al. 2017); regression for observed SPD matrix outcomes, e.g., diffu-

Psychophysiological interactions (PPI)

A **psychophysiological interaction (PPI)** is the product of the task ("psychological") and a seed region's BOLD signal ("physiological") (Friston et al. 1997, McLaren et al. 2012). For a chosen seed v', at each region v:

$$y_{it,v} = \theta_{0,iv} + \theta_{1,iv}\, y_{it,v'} + \beta_{iv}\, x_t + \alpha_{i,vv'}\left(x_t \cdot y_{it,v'}\right) + \text{nuisance} + e_{it,v},$$

- $y_{it,v}$: BOLD response at region v; $\theta_{1,iv}$: task-independent coupling with the seed (seed-based connectivity); β_{iv}: task activation.
- $\alpha_{i,vv'}$: task-induced change in the (v, v') association — cf. $\boldsymbol{\alpha}_j$ with $J = 1$ in the matrix-log model, but pairwise and asymmetric: the $\hat{\alpha}_{i,vv'}$ do not form a valid covariance matrix.
- Group analysis: two-stage summary statistics — massive-univariate second-level tests on the $\hat{\alpha}_{i,vv'}$ across participants.
- Poor reliability: ICC = 0.07 in a visual checkerboard task (Di and Biswal 2017).

Covariance regression for task fMRI

For each child $i = 1, \ldots, n$: first-level mean model across v regions: $\boldsymbol{y}_{it} = \boldsymbol{X}_t \boldsymbol{\beta}_i + \boldsymbol{Z}_{it} \boldsymbol{\delta}_i + \boldsymbol{e}_{it}$. Model the **time-varying error covariance**:

$$\boldsymbol{\Sigma}_{it} = \mathrm{Cov}(\boldsymbol{e}_{it}) = \exp(\boldsymbol{\eta}_{it}), \qquad \boldsymbol{\eta}_{it} = \sum_{j=1}^{J} x_{ijt} \boldsymbol{\alpha}_j, \qquad x_{i1t} = 1.$$

- Our application: x_{i2t}: movie-block predictor (boxcar $*$ HRF) — does the task change the covariance between brain regions?
- $\boldsymbol{\alpha}_2 \in \mathbb{R}^{V \times V}$, symmetric: the task effect on each entry of the log covariance matrix — the estimand.
- Challenges: the errors $\boldsymbol{e}_{it}$ are unobserved — we only have residuals $\boldsymbol{r}_{it}$; temporal dependence within child; population inference $\Rightarrow$ generalized estimating equations (GEE).

Novel approach: covreg sandwich

We propose a GEE approach to covariance regression with clustered data.

- For now, errors are observed: $\boldsymbol{z}_{it} = \text{vech}(\boldsymbol{e}_{it}\boldsymbol{e}_{it}^\top)$, $\text{E}(\boldsymbol{z}_{it}) = \text{vech}(\boldsymbol{\Sigma}_{it}) = \boldsymbol{\sigma}_{it}(\boldsymbol{\alpha})$, stacked over time as $\boldsymbol{z}_i, \boldsymbol{\sigma}_i \in \mathbb{R}^{TQ}$, $Q = V(V+1)/2$.
- GEE for $\boldsymbol{\alpha} = \{\text{vech}(\boldsymbol{\alpha}_1)^\top, \ldots, \text{vech}(\boldsymbol{\alpha}_J)^\top\}^\top$, solved by IRLS:

$$\boldsymbol{U}(\boldsymbol{\alpha}) = \sum_{i=1}^{n} \boldsymbol{D}_i^\top \boldsymbol{V}_i^{-1} \{\boldsymbol{z}_i - \boldsymbol{\sigma}_i(\boldsymbol{\alpha})\} = \boldsymbol{0}, \qquad \boldsymbol{D}_i = \partial\boldsymbol{\sigma}_i/\partial\boldsymbol{\alpha}^\top. \tag{1}$$

- Working covariance: $\boldsymbol{V}_i = \boldsymbol{A}_i^{1/2}\{\boldsymbol{R}_i(\hat{\phi}^2) \otimes \boldsymbol{I}_Q\}\boldsymbol{A}_i^{1/2}$ — Gaussian fourth-moment blocks $\boldsymbol{A}_{it}$ (Prentice and Zhao 1991) with edge-specific variances (MOM); Gaussian AR(1) errors ⇒ the product residuals have serial correlation ϕ^2, so the working correlation is *fixed* at $\hat{\phi}^2$.
- Sandwich variance estimators (Liang and Zeger 1986):

$$\widehat{\text{Cov}}(\hat{\boldsymbol{\alpha}}) = \left(\textstyle\sum_i \boldsymbol{D}_i^\top \boldsymbol{V}_i^{-1} \boldsymbol{D}_i\right)^{-1} \left(\textstyle\sum_i \boldsymbol{D}_i^\top \boldsymbol{V}_i^{-1} \{\boldsymbol{z}_i - \boldsymbol{\sigma}_i\}\{\boldsymbol{z}_i - \boldsymbol{\sigma}_i\}^\top \boldsymbol{V}_i^{-1} \boldsymbol{D}_i\right) \left(\textstyle\sum_i \boldsymbol{D}_i^\top \boldsymbol{V}_i^{-1} \boldsymbol{D}_i\right)^{-1}$$

Insight into D_i: differentiating the matrix exponential

1. $\boldsymbol{\alpha} \mapsto \boldsymbol{\sigma}_{it}$ is a composition of a linear map and the matrix exponential:

$$\boldsymbol{\alpha} \xrightarrow{\text{linear}} \operatorname{vech}(\boldsymbol{\eta}_{it}) = \boldsymbol{X}_{it}\,\boldsymbol{\alpha} \xrightarrow{\exp} \boldsymbol{\sigma}_{it} = \operatorname{vech}\big(\exp(\boldsymbol{\eta}_{it})\big), \qquad \boldsymbol{X}_{it} = \boldsymbol{x}_{it}^{\top} \otimes \boldsymbol{I}_Q.$$

2. Chain rule — differentiate each map and multiply:

$$\frac{\partial \boldsymbol{\sigma}_{it}}{\partial \boldsymbol{\alpha}^{\top}} = \underbrace{\frac{\partial \operatorname{vech}(\boldsymbol{\Sigma}_{it})}{\partial \operatorname{vech}(\boldsymbol{\eta}_{it})^{\top}}}_{\boldsymbol{J}_{it} \in \mathbb{R}^{Q \times Q}} \underbrace{\frac{\partial \operatorname{vech}(\boldsymbol{\eta}_{it})}{\partial \boldsymbol{\alpha}^{\top}}}_{\boldsymbol{X}_{it}}. \tag{2}$$

3. The Fréchet derivative of the matrix exponential (Daleckii and Krein 1965, Higham 2008): with $\boldsymbol{\eta}_{it} = \boldsymbol{U}_{it} \operatorname{diag}(\boldsymbol{\lambda}_{it})\, \boldsymbol{U}_{it}^{\top}$, the b-th column of $\boldsymbol{J}_{it}$ is

$$[\boldsymbol{J}_{it}]_b = \operatorname{vech}\Big[\boldsymbol{U}_{it}\big\{\boldsymbol{\Psi}_{it} \odot \big(\boldsymbol{U}_{it}^{\top} \boldsymbol{E}_b\, \boldsymbol{U}_{it}\big)\big\}\boldsymbol{U}_{it}^{\top}\Big], \qquad [\boldsymbol{\Psi}_{it}]_{kk'} = \frac{e^{\lambda_{it,k}} - e^{\lambda_{it,k'}}}{\lambda_{it,k} - \lambda_{it,k'}},$$

where $\boldsymbol{E}_b$ has ones at the b-th vech coordinate's positions (v, v') and (v', v), zero elsewhere; $[\boldsymbol{\Psi}_{it}]_{kk'} = e^{\lambda_{it,k}}$ when $\lambda_{it,k} = \lambda_{it,k'}$; $\odot$: entrywise product.

4. Insight: the matrix version of the scalar log-linear variance model, $\log\sigma^2 = x\beta \Rightarrow \partial\sigma^2/\partial\beta = x\, e^{x\beta}$: here $\boldsymbol{X}_{it} \leftrightarrow x$ and $\boldsymbol{J}_{it} \leftrightarrow e^{x\beta}$.

Leverage adjustment: residuals are not errors

The first-level fit leaves residuals, not errors: with the (GLS) hat matrix $\boldsymbol{H}_i(\hat{\phi})$ of the mean-model design $\boldsymbol{S}_i$, stacking time-major,

$$\boldsymbol{r}_i = \left\{(\boldsymbol{I}_T - \boldsymbol{H}_i(\hat{\phi})) \otimes \boldsymbol{I}_V\right\} \boldsymbol{e}_i.$$

$$\mathrm{E}(\boldsymbol{r}_i \boldsymbol{r}_i^\top) = \left\{(\boldsymbol{I}_T - \boldsymbol{H}_i) \otimes \boldsymbol{I}_V\right\} \boldsymbol{\Omega}_i \left\{(\boldsymbol{I}_T - \boldsymbol{H}_i)^\top \otimes \boldsymbol{I}_V\right\} = \widetilde{\boldsymbol{\Sigma}}_i, \qquad \mathrm{E}(\boldsymbol{r}_{it} \boldsymbol{r}_{it}^\top) = \widetilde{\boldsymbol{\Sigma}}_{i,tt} \neq \boldsymbol{\Sigma}_{it}. \tag{3}$$

- $\boldsymbol{\Omega}_i = \mathrm{E}(\boldsymbol{e}_i \boldsymbol{e}_i^\top)$: diagonal blocks are $\boldsymbol{\Sigma}_{i,tt}$.
  Off-diagonal blocks are $\phi^{|t-t'|} \boldsymbol{\Sigma}_{it}^{1/2} \boldsymbol{\Sigma}_{it'}^{1/2}$.
- Leverage adjustment: in (1), replace $\boldsymbol{\sigma}_{it}$ with $\widetilde{\boldsymbol{\sigma}}_{it} = \mathrm{vech}(\mathrm{E}(\boldsymbol{r}_{it} \boldsymbol{r}_{it}^\top))$.

Two-stage estimation algorithm

Stage 1: iterated feasible GLS and AR(1) estimation (maximum lag $L = 20$)

Initialize $\hat{\phi}^{(0)} = 0$.
repeat
  For each child: update the GLS mean coefficients given $\hat{\phi}^{(m)}$; form residuals $\boldsymbol{r}_{it}^{(m)}$ via the hat matrix $\boldsymbol{H}_i(\hat{\phi}^{(m)})$.
  Worsley bias-reduction step (Worsley et al. 2002): estimate node-wise residual autocovariances $\hat{\boldsymbol{\gamma}}_{iv}^{(m)}$, correcting the bias induced by the mean-model fit.
  Pooled Yule–Walker update: $\hat{\phi}^{(m+1)} = \sum_{i,v} \hat{\gamma}_{iv}^{(m)}(1) / \sum_{i,v} \hat{\gamma}_{iv}^{(m)}(0)$.
until $\hat{\phi}$ and the mean coefficients converge

Stage 2: adjusted covariance regression, working correlation fixed at $\hat{\phi}^2$

Initialize $\hat{\boldsymbol{\alpha}}^{(0)}$ and edge dispersions $\hat{\boldsymbol{\tau}}^{(0)}$.
repeat
  Product residuals minus their expectation: $\boldsymbol{z}_{it} - \widetilde{\boldsymbol{\sigma}}_{it}(\hat{\boldsymbol{\alpha}}^{(s)})$, with $\widetilde{\boldsymbol{\sigma}}_{it}$ from (3); derivatives $\widetilde{\boldsymbol{D}}_i^{(s)}$ from (2).
  Pseudo-observations: $\boldsymbol{y}_i^{(s)} = \widetilde{\boldsymbol{D}}_i^{(s)} \hat{\boldsymbol{\alpha}}^{(s)} + \boldsymbol{z}_i - \widetilde{\boldsymbol{\sigma}}_i^{(s)}$.
  Fisher scoring = weighted least squares on the pseudo-observations: $\hat{\boldsymbol{\alpha}}^{(s+1)} = \left(\sum_i \widetilde{\boldsymbol{D}}_i^\top \boldsymbol{V}_i^{-1} \widetilde{\boldsymbol{D}}_i\right)^{-1} \sum_i \widetilde{\boldsymbol{D}}_i^\top \boldsymbol{V}_i^{-1} \boldsymbol{y}_i^{(s)}$; update $\hat{\boldsymbol{\tau}}^{(s+1)}$ by Pearson moments.
until $\hat{\boldsymbol{\alpha}}$ and $\hat{\boldsymbol{\tau}}$ converge
Compute the sandwich variance of $\hat{\boldsymbol{\alpha}}$.

Simulation design

Designed to mimic the real data: $n = 36$ children, $T = 304$ volumes, $V = 30$ regions, 200 replicates.

- Data generation: $\boldsymbol{y}_{it} = \boldsymbol{X}_t \boldsymbol{\beta}_i + \boldsymbol{Z}_{it} \boldsymbol{\delta}_i + \boldsymbol{\Sigma}_{it}^{1/2} \boldsymbol{w}_{it}$, node-wise latent AR(1) $\boldsymbol{w}_{it}$ with $\phi = 0.6$; subject-specific mean coefficients $\boldsymbol{\gamma}_i \sim N(0,1)$ entrywise.
- Truth: $\boldsymbol{\Sigma}_{it} = \exp(\boldsymbol{\alpha}_1 + x_{i2t}\, \boldsymbol{\alpha}_2)$; $\boldsymbol{\alpha}_1$ has a seven-network block structure (stochastic block model, log-eigenvalues in $(-0.3, 0.3)$); $\boldsymbol{\alpha}_2 = \mathbf{0}$: the movie effect is **null** $\Rightarrow$ evaluate type I error and coverage.
- Designs from the real data: the movie predictor x_{i2t} (boxcar $*$ HRF, standardized) enters twice, as in the real analysis — in the mean model and the covariance model.
- Compare two estimators of the $Q \times 2 = 930$ parameters:
  1. raw two-step: residual products $\mathrm{vech}(\boldsymbol{r}_{it} \boldsymbol{r}_{it}^\top)$ treated as if they were error products;
  2. leverage-adjusted covariance: mean $\tilde{\boldsymbol{\sigma}}_{it} = \mathrm{vech}(\tilde{\boldsymbol{\Sigma}}_{i,tt})$.

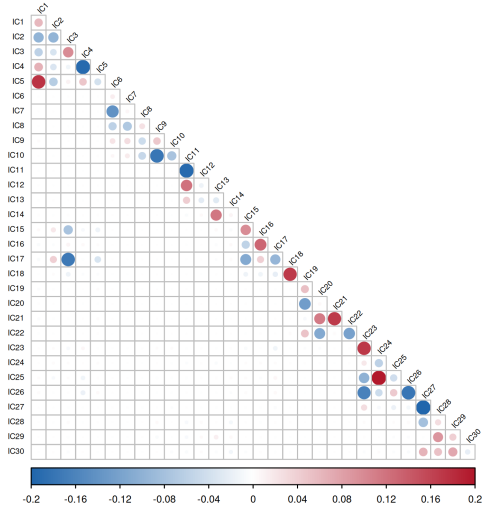
Simulation results: the movie effect α_2 (truth = $\mathbf{0}$)

Parameters	Method	Bias	RMSE	Type I error
Off-diagonal (435)	Raw two-step	0.0000	0.0131	0.074
	Adjusted score	0.0000	0.0144	0.070
Diagonal (30)	Raw two-step	0.0447	0.0483	0.695
	Adjusted score	0.0067	0.0214	0.085

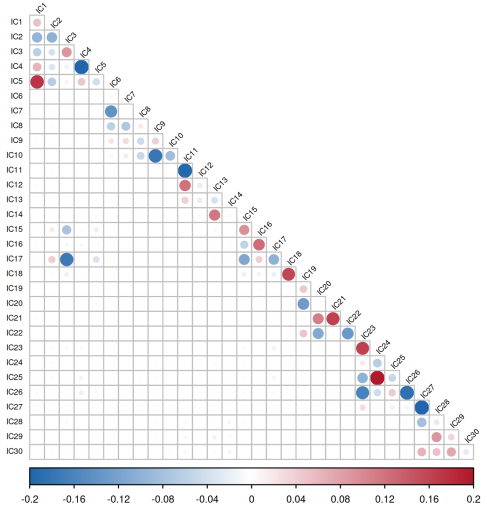
- Off-diagonal coefficients: both methods essentially unbiased with type I error near the nominal 0.05.
- Diagonal (log-variance) coefficients: ignoring the leverage biases the raw two-step estimator upward (0.045, several empirical SDs), inflating the type I error to 0.70.
- The leverage adjustment decreases the bias; type I error is 0.085.

Simulation results: coefficient recovery (adjusted score)

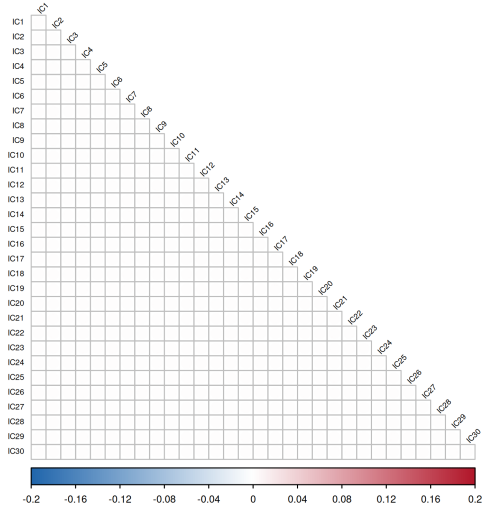
True α_1



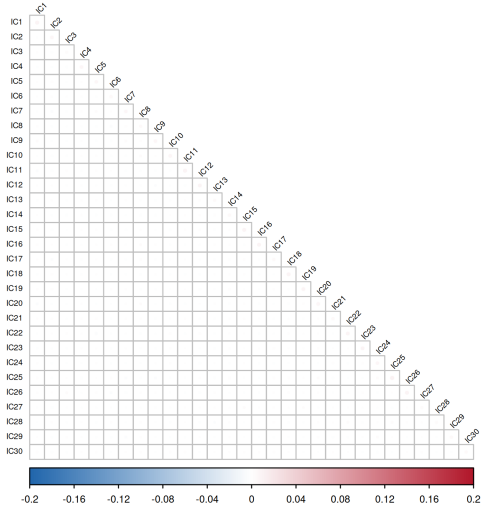
Mean estimated α_1 (200 replicates)



True α_2



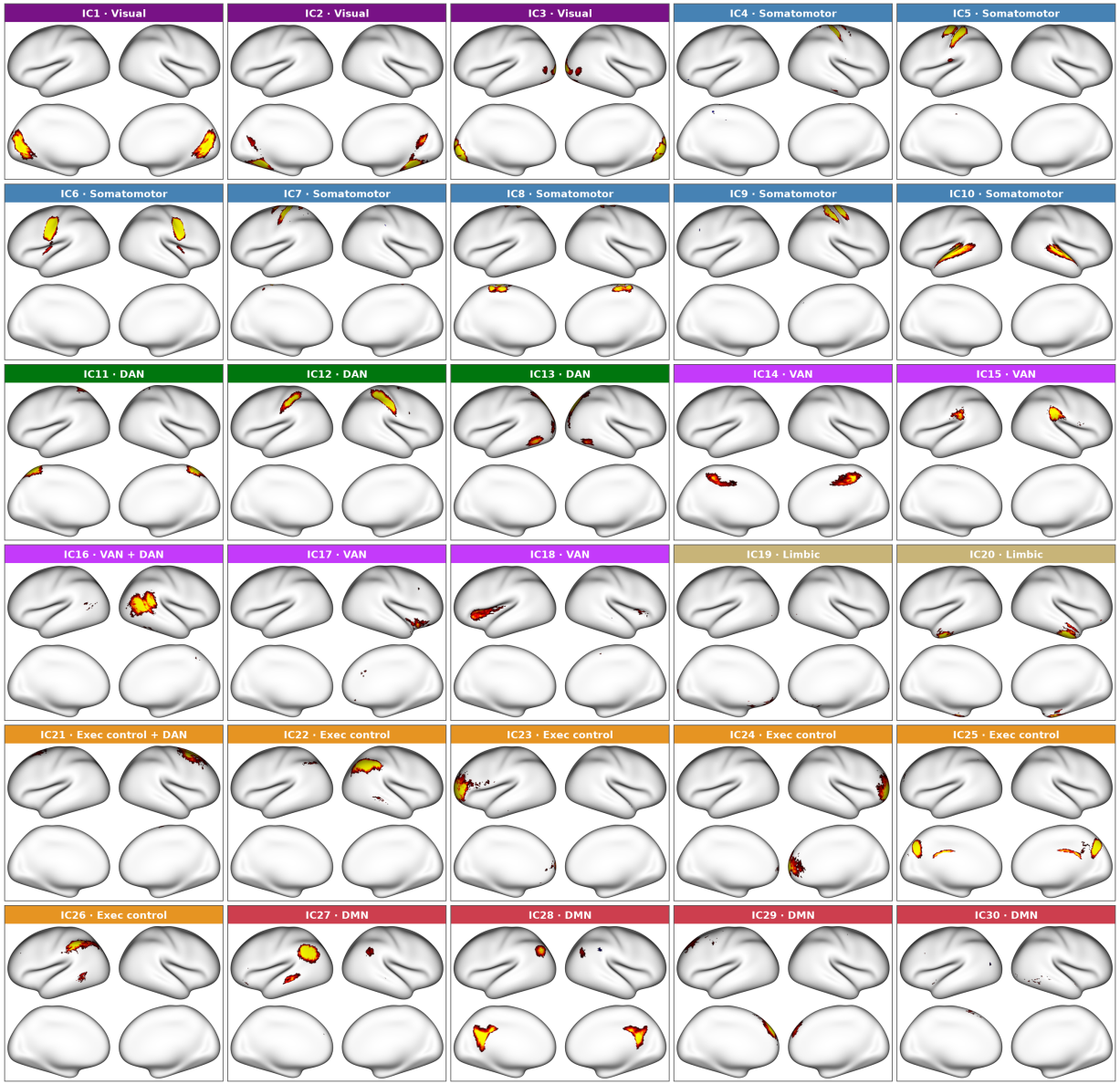
Mean estimated α_2 (200 replicates)



Dimension reduction with Sparse ICA

- We apply covreg sandwich to the movie-watching task.
- For analytic and interpretation feasibility, we reduce the dimension of the data from ~90,000 CIFTI vertices to $V = 30$ components estimated using Sparse ICA (Wang et al., Risk, 2024).
- The components serve the same role as a weighted brain parcellation. Here, they are study- and task-specific.
- We project each child's task data onto the 30 group Sparse ICA components, reducing the cortical surface fMRI data from $90{,}000 \times 304$ to 30×304.
- Each component is assigned to a Yeo-7 resting-state network by the share of its squared loadings in each network.

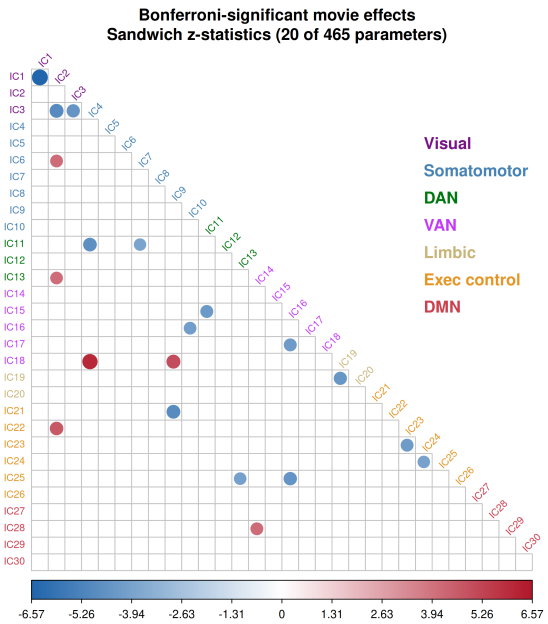
Group Sparse ICA components



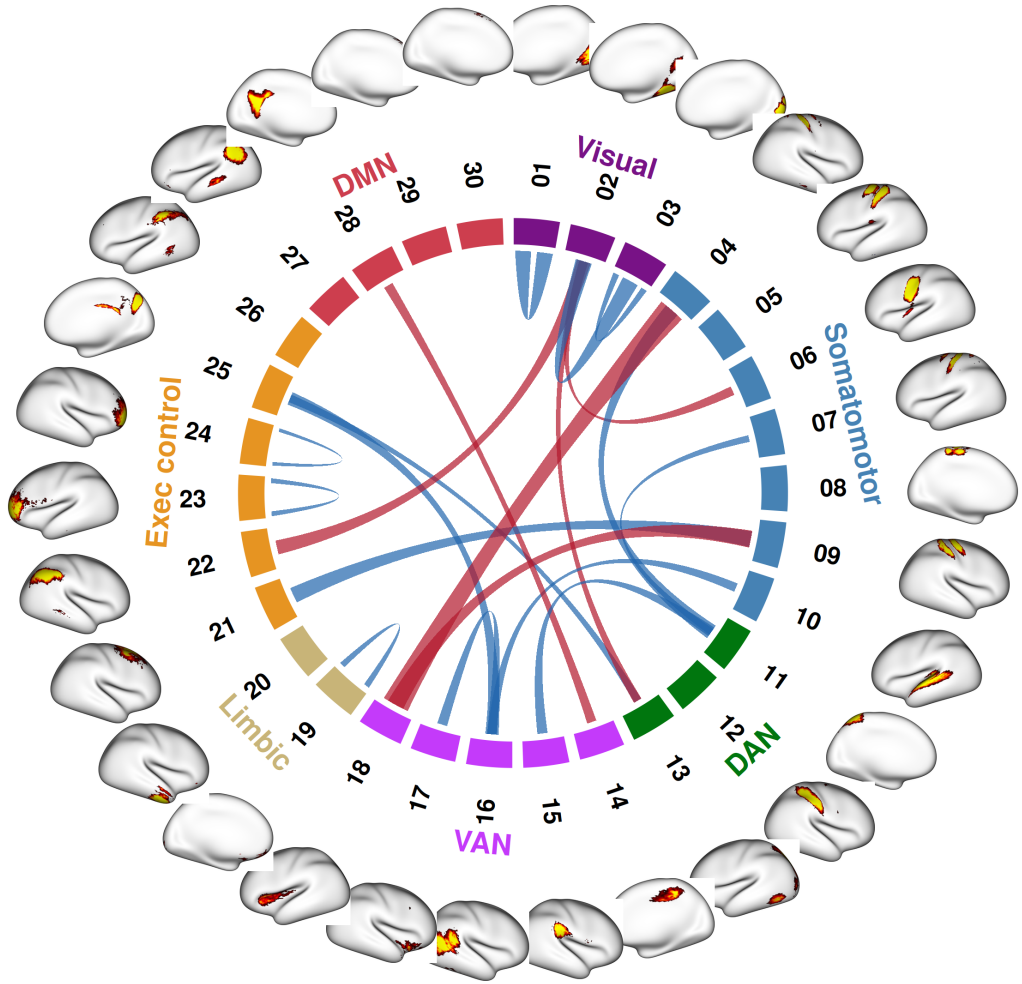
Covariance regression of movie-watching connectivity

- We fit the first-level task activation model previously described, then extract the residuals.
- We fit covreg sandwich with the log-Euclidean link (leverage-adjusted score).
- Here, we present the results on the **log-Euclidean scale**.
- Note: edges that are significant on the log-Euclidean scale in general do not correspond to edges significant on the positive-definite scale. Future work will investigate this.

Movie-watching effects on connectivity (Bonferroni)



20 of 465 parameters significant (15 connections, 5 log-variances). Red: increase during movie blocks; blue: decrease.



Interpretation

- Ventral attention / salience couples with somatomotor cortex: the two largest positive effects are IC18–IC4 ($z = 6.1$) and IC18–IC9 ($z = 5.0$); VAN–DMN (IC14–IC28) also increases.
- Attention and control networks decouple: negative effects on DAN–somatomotor, executive–somatomotor, VAN–DAN, executive–DAN, and executive–VAN connections.
- Reduced component variances: the movie decreases the log-variance of visual (IC1, IC3), limbic (IC19), and executive control (IC23, IC24) components; the largest effect in the analysis is primary visual (IC1, $z = -6.6$).
- Visual cortex integrates with the rest of the brain: visual components couple more strongly with somatomotor (IC6), dorsal attention (IC13), and executive control (IC22) components, while within-visual covariance decreases (IC2–IC3).
- Effects are entries of $\log \boldsymbol{\Sigma}_{it}$ (log-Euclidean scale), not correlations.

Discussion

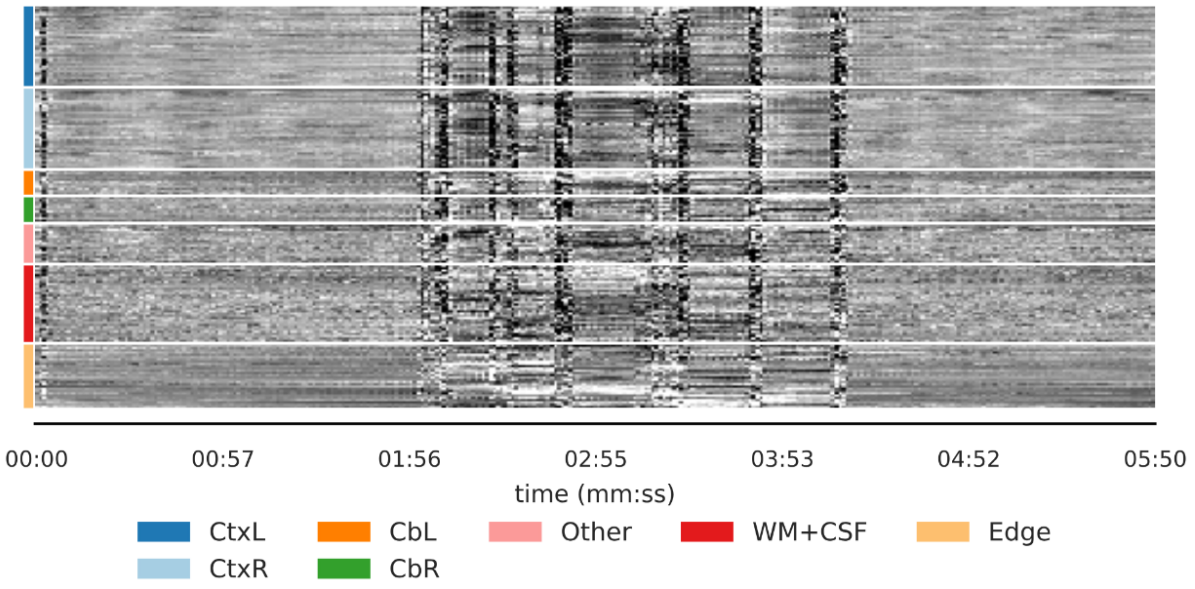
- Covariance regression provides a formal framework for how a task modulates connectivity: the effect on the full $V \times V$ covariance matrix, respecting positive definiteness.
- covreg sandwich: a GEE approach with sandwich variance estimators for clustered fMRI time series; the leverage adjustment accounts for using first-level residuals in place of errors (bias 0.045 → 0.007, type I error 0.70 → 0.085).
- Results: movie watching reorganizes connectivity in children — ventral attention/salience couples with somatomotor, attention and control networks decouple, and visual variances decrease while visual components integrate with other networks.
- Future work: Examine how task connectivity patterns differ in autism. Examine eye-tracking data (e.g., eye fixations) as a measure of attention.

Motion quality control creates selection bias

Brain connectivity and autism

- Autism is a neurodevelopmental disorder thought to be related to alterations in brain function.
- Resting-state fMRI is used to examine functional connectivity, or correlations between brain regions.
- Connectivity theory of autism:
  - Underconnectivity in long-range connections (Just et al. 2012).
  - Overconnectivity in local connections (Keown et al. 2013).
- Disruptions in the default mode network (Yerys et al. 2015), including anterior-to-posterior parts (Di Martino et al. 2013).
- Some similarities with the patterns produced by motion artifacts (Deen and Pelphrey 2012).

Problem I: Motion Artifacts



Rigid body motion correction

↓

Nuisance signals regression (Ciric et al. 2017)

↓

Remove scans with excessive motion (Satterthwaite et al. 2013, Power 2017, Parkes et al. 2018)

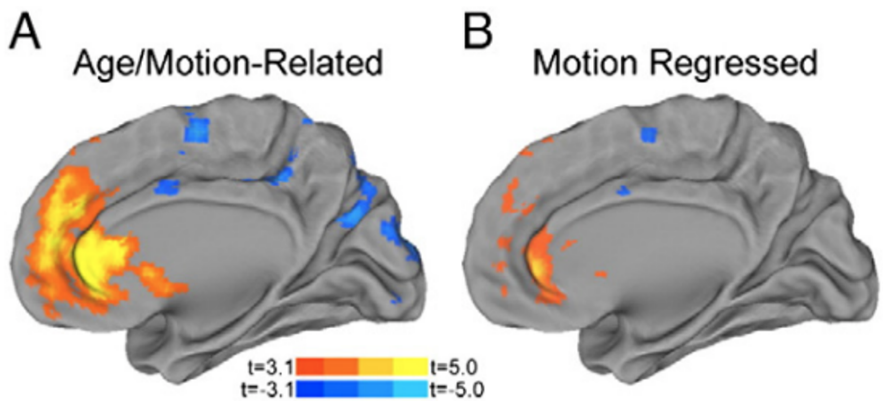
- Existing motion control using regression is inadequate because motion patterns are complex (Power et al. 2014).
- Current best practices remove time points with framewise displacement >0.2 mm, then remove participants if < 5 minutes of data remain after this scrubbing.

Challenges in pediatric neuroimaging

Motion challenges:

- Younger children move more (Satterthwaite et al. 2012)
- Confounding with neurodevelopmental trajectories.

"It really, really, really sucks. My favorite result of the last five years is an artifact." -*Steven Petersen, Professor of Cognitive Neuroscience at Washington University in St. Louis.*



Default mode decreased connectivity

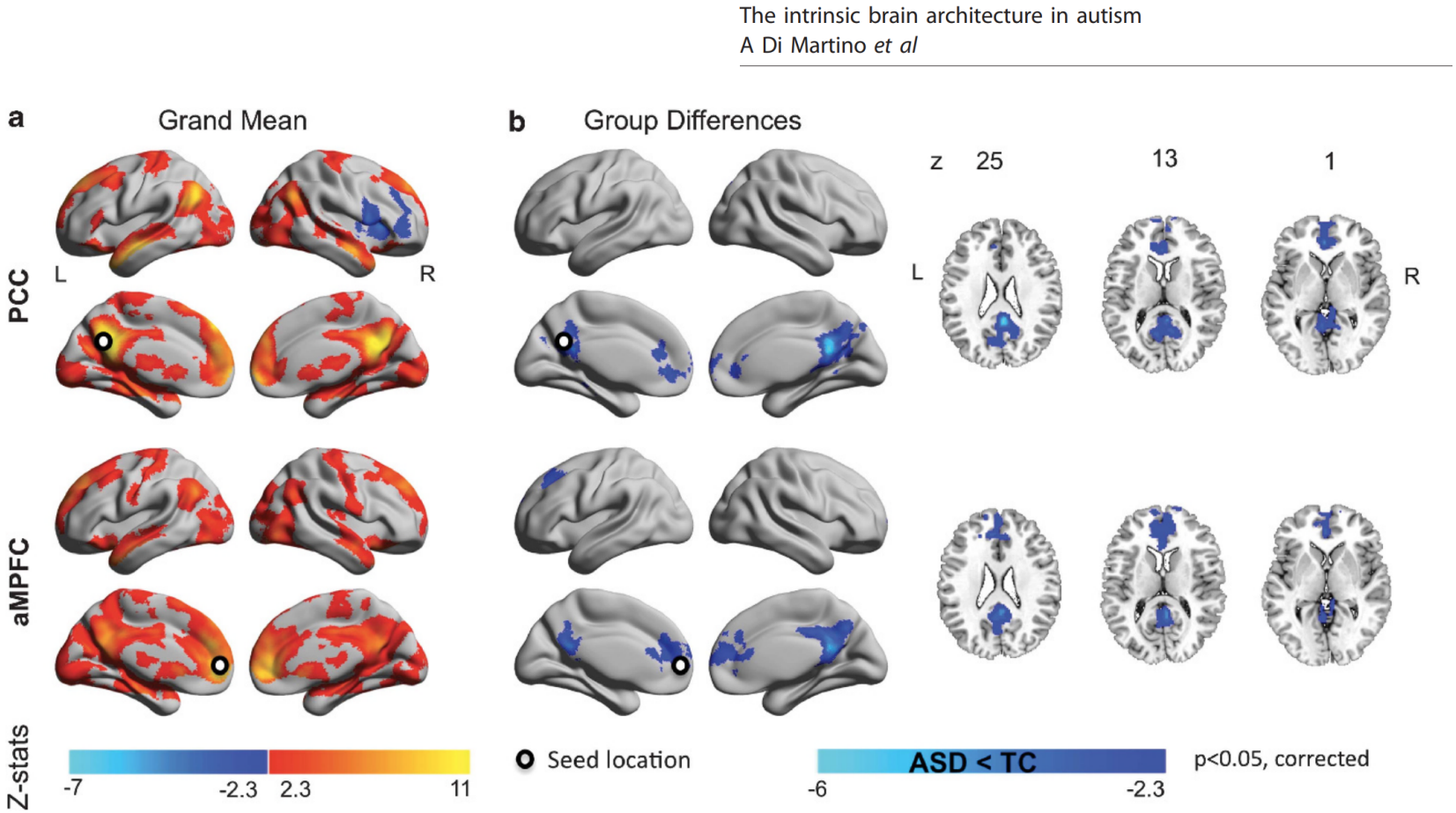
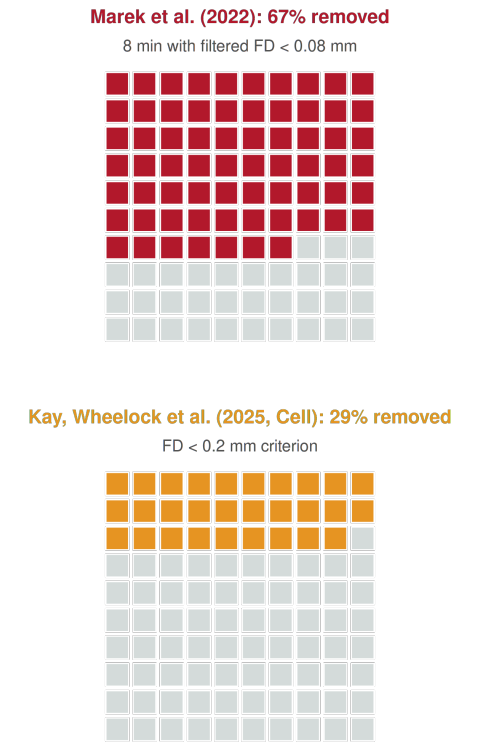


Figure: Di Martino et al. (2014): excluded high-motion participants, covaried mean FD; robust to scrubbing.

Problem II: Motion QC creates selection bias

- State of the art: remove high motion participants.
- Drastic reductions in sample size: the ABCD study **removed 67%** of children (Marek et al. 2022); recent criteria are less restrictive but still remove 29% (Kay et al. 2025).
- In ABCD, this creates selection bias, disproportionately selecting for: higher SES, White participants, older, females, higher neurocognitive skills, fewer neurodevelopmental problems (Cosgrove et al. 2022).
- Across the lifespan, exclusion disproportionately removes younger and older participants, higher BMI, and motion-associated clinical conditions (Sadil et al. 2026).
- Unethical?



100 squares = 100 children

Participant removal creates selection bias in ASD

M.B. Nebel, D.E. Lidstone, L. Wang et al. NeuroImage 257 (2022) 119296

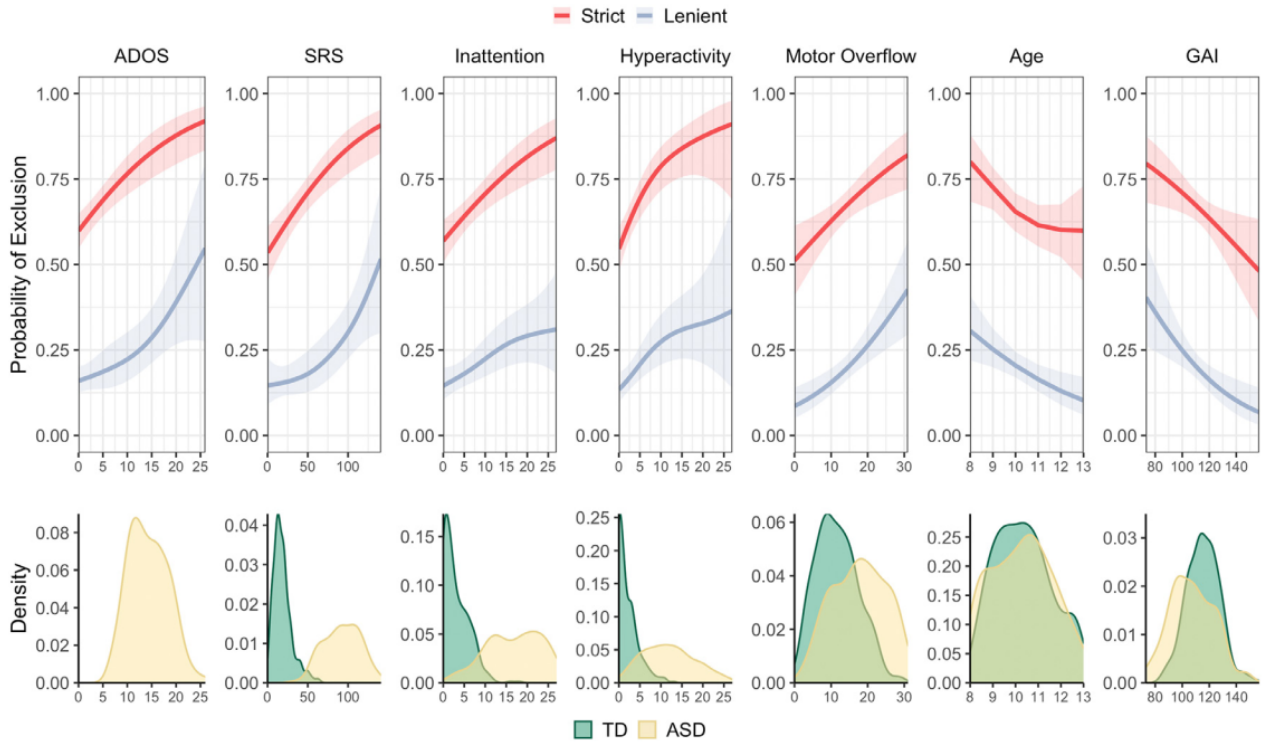


Fig. 4. rs-fMRI exclusion probability changes with phenotype and age. Univariate analysis of rs-fMRI exclusion probability as a function of participant characteristics. From left to right: Autism Diagnostic Observation Schedule (ADOS) total scores, social responsiveness scale (SRS) scores, inattentive symptoms, hyperactive/impulsive symptoms, total motor overflow, age, and general ability index (GAI) using the lenient (slate blue lines, all FDR-adjusted $p < 0.01$), and strict (red lines) motion quality control (all FDR-adjusted $p < 0.03$). Variable distributions for each diagnosis group (included and excluded scans) are displayed across the bottom panel (TD=typically developing, green; ASD=autism spectrum disorder, yellow).

Figure: Nebel et al., Risk (2022)

Motion increases with autism symptoms

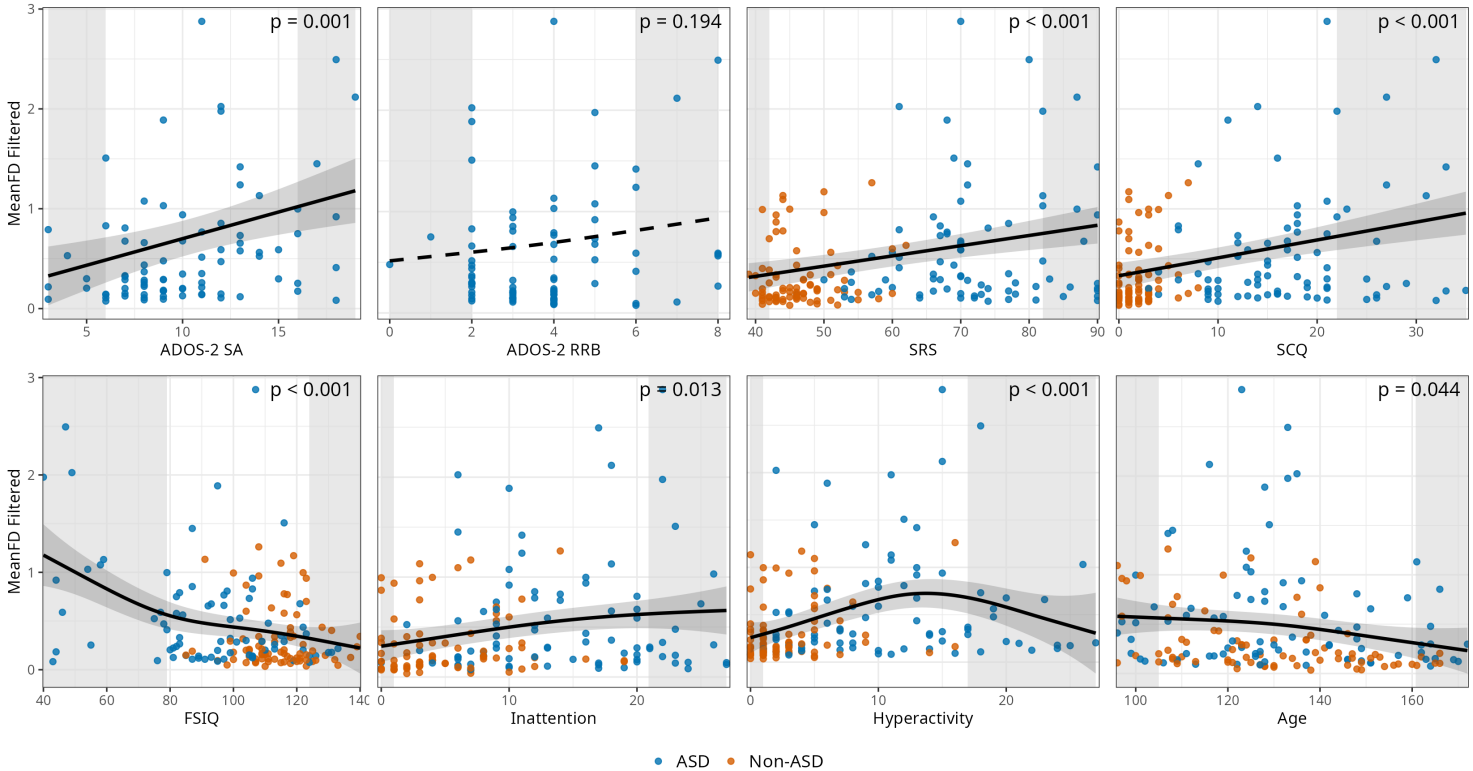


Figure: GAMs of mean filtered framewise displacement on participant characteristics, adjusting for age and sex (ASD: blue; non-ASD: orange). Dashed curves: smooth term $p \geq 0.05$. Results from the Emory University Brain Connectivity Study of 163 children (autistic and non-ASD) with Dr. Sarah Shultz.

Motion exclusion probability increases with autism symptoms

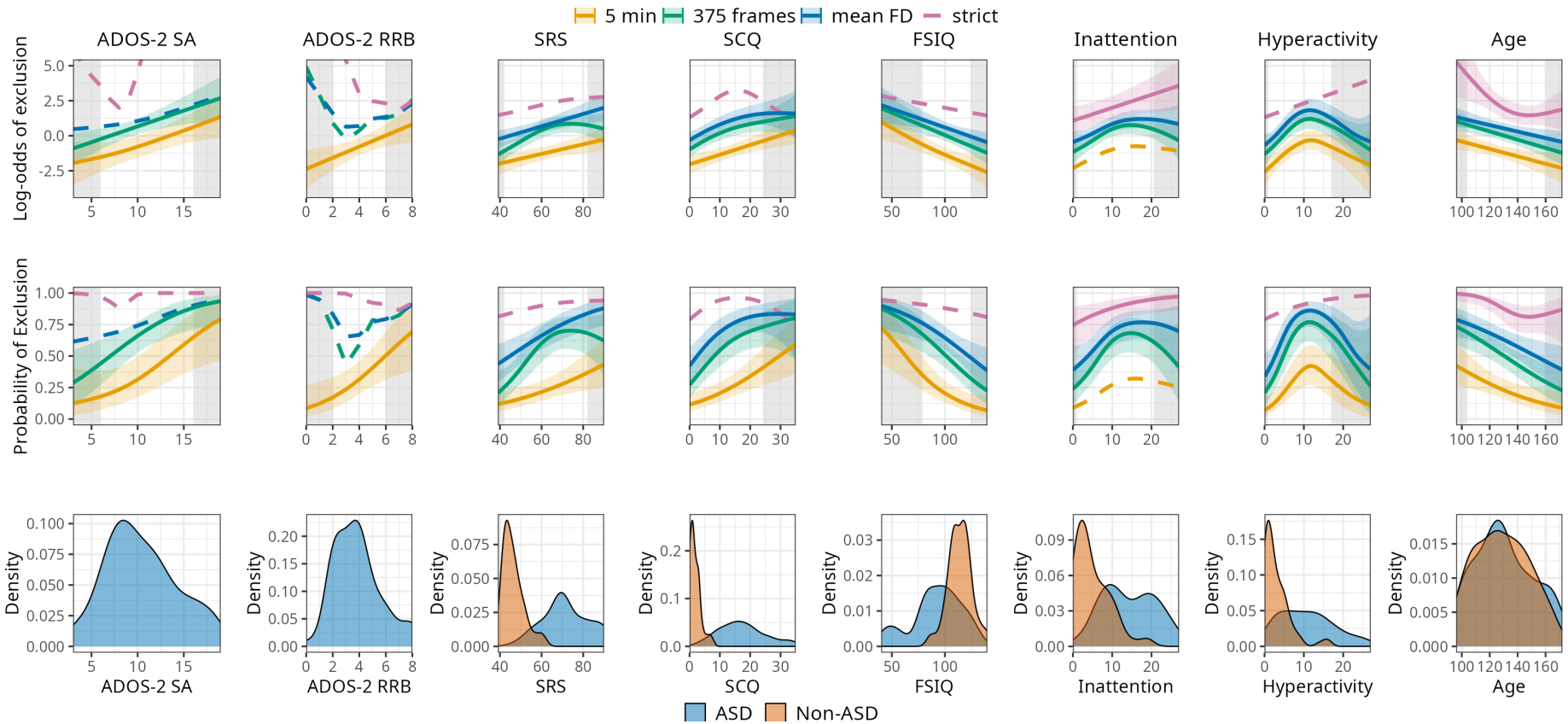


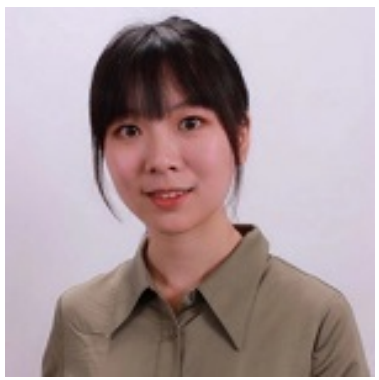
Figure: Probability of rs-fMRI exclusion as a function of participant characteristics under four motion QC criteria, adjusting for age and sex. From the Emory University Brain Connectivity Study.

Motion Control (MoCo) in Functional Connectivity Studies in Children with Autism Spectrum Disorder

Jialu Ran[1], Sarah Shultz[2], Benjamin Risk[1] and David Benkeser[1]

[1] *Department of Biostatistics and Bioinformatics,* [2] *Department of Pediatrics, Emory University*

Ran, J., Shultz, S., Risk, B. B., and Benkeser, D. (2026). Nonparametric motion control in functional connectivity studies in children with autism spectrum disorder. *Biometrics*, 82(2).

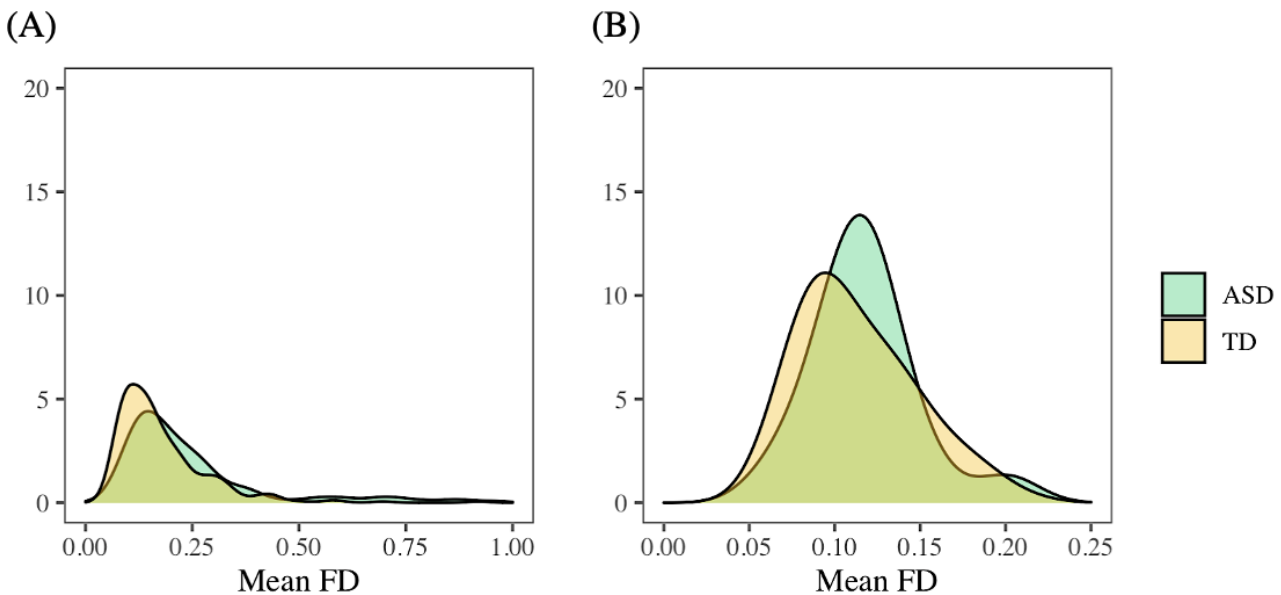


Stochastic Intervention

What would the autistic brain look like under low motion?

We use the framework of stochastic intervention (Díaz et al. 2021).

Replace mean framewise displacement M by tolerable value
$M \sim P_{M|\Delta=1,A=0,X}$



Motion-Controlled Estimand (MoCo)

Motion-Controlled Estimand (MoCo): the difference between the autistic and non-ASD brain when all children have a tolerable amount of motion.

- A: non-ASD (0), ASD (1)
- M: a motion variable (mean framewise displacement)
- X: demographic variables.
- Z: variables related to autism symptoms.

$$\int \Big[\big\{\mu_{Y|A,M,X,Z}(1,m,x,z)\,p_{Z|A,X}(z|1,x) - \mu_{Y|A,M,X,Z}(0,m,x,z)\,p_{Z|A,X}(z|0,x)\big\}\, p_{M|\Delta=1,A,X}(m|0,x)\,p_X(x)\Big]\,dz\,dm\,dx\,.$$

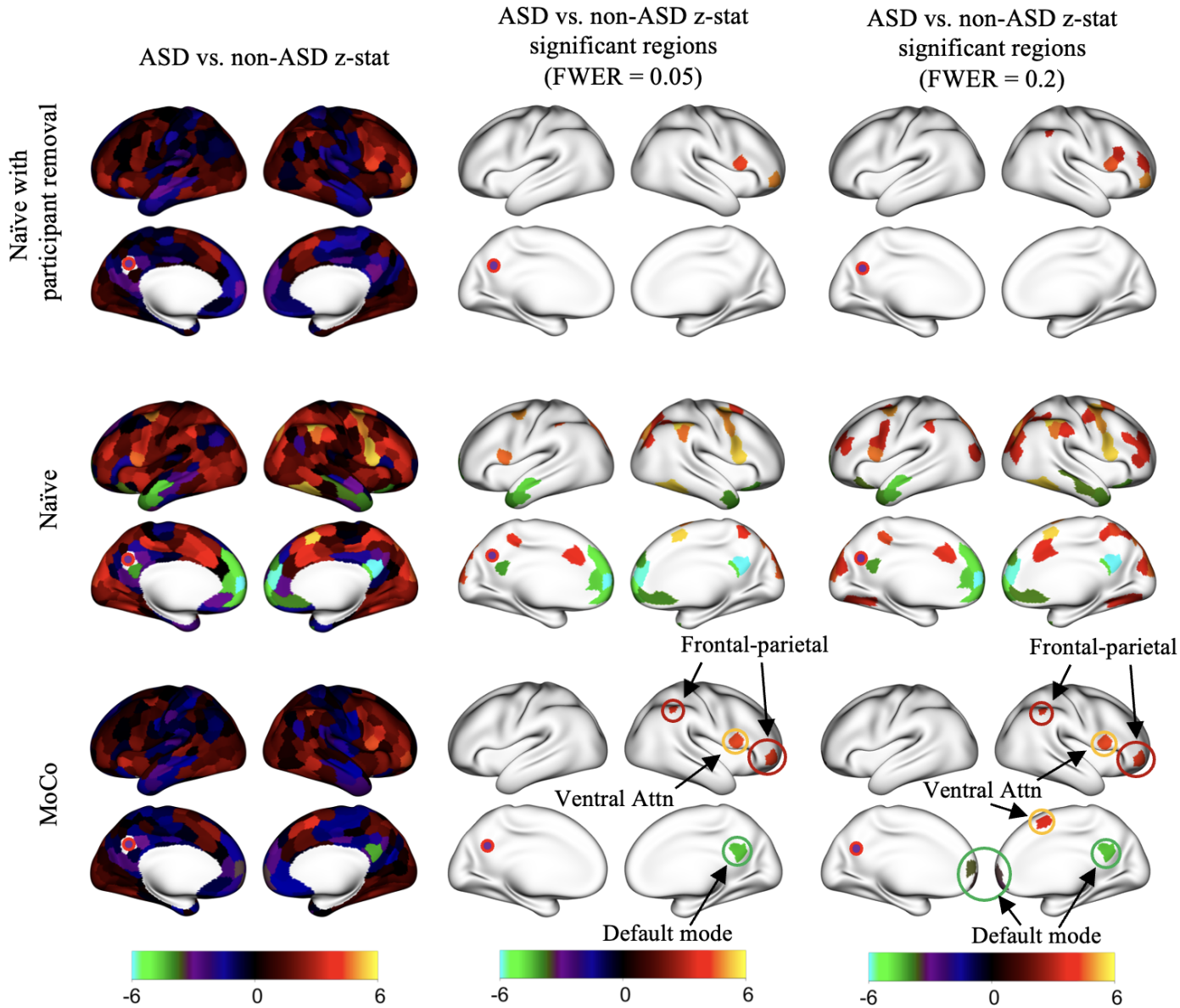
Intuition: estimating what brain looks like under the counterfactual of low motion.

ABIDE Data

School-age children from Autism Brain Imaging Data Exchange (ABIDEI and ABIDEII) Dataset (Di Martino et al. 2014; 2017)

- variables:
  - A_i: 245 TD ($A = 0$), 132 ASD ($A = 1$) [377 8-13 yo children].
  - X_i: age, sex, handedness.
  - Z_i: autism diagnostic observation schedule, IQ, medication status.
  - M_i: mean frame-wise displacement (FD).
  - $\Delta_i = 1$: > 5 minutes of data free from ≥ 0.2 framewise displacement (Power et al. 2014) [126 TD (51%), 34 ASD (26%)].
- Y_{ij}: correlation between seed region in DMN and region j, $j = 1, \ldots, 400$ (Schaeffer 400 atlas).
- SuperLearner for nuisance regressions: random forest, xgboost, multivariate adaptive regression splines, LASSO, ridge regression, generalized additive models, generalized linear models (with and without interactions, and with and without forward stepwise covariate selection)
- Highly adaptive lasso for conditional density estimation (Hejazi et al. 2022).
- Cross-fitting with 5-fold cross-validation.

ABIDE Inference



MoCo software

Input

vector of length n:

- diagnosis (A)
- motion (M)
- motion inclusion indicator (Δ_M)
- brain image quality indicator (Δ_Y)

vector of length n or data frame of size n x p:

- demographic confounders (X)
- behavior phenotypes (Z)

matrix of size n x p:

- outcome matrix (Y), rows are participants and columns are brain phenotypes across regions

⟹

Output

vector of length n:

- motion-controlled brain phenotypes for each group
- motion-controlled group difference in brain phenotypes
- z scores
- significant brain phenotypes

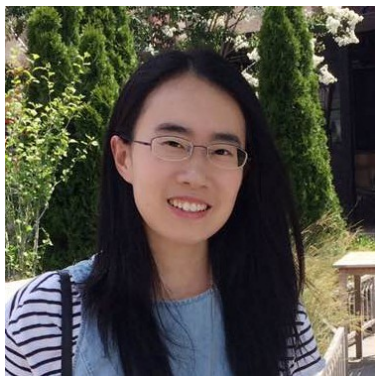


R package: https://github.com/thebrisklab/MoCo

Analysis of ADHD in the ABCD dataset using MoCo

Xiyan Tan and Benjamin Risk

Department of Biostatistics and Bioinformatics, Emory University



Selection bias in ABCD: demographics (X)

ABCD 7.0 baseline, $n = 7{,}393$ children with observed FC; motion exclusion removes 1,577 (21.3%).

	Non-ADHD ($n = 6{,}848$)			ADHD ($n = 545$)		
Demographic (X)	Excluded	Included	p	Excluded	Included	p
Age, mean (SD)	9.84 (0.59)	9.97 (0.62)	<0.001	9.92 (0.63)	9.96 (0.63)	0.423
Male	820 (58.7%)	2577 (47.3%)	<0.001	131 (73.2%)	251 (68.6%)	0.270
Race: Black	233 (16.7%)	652 (12.0%)	<0.001	31 (17.3%)	53 (14.5%)	0.584
Hispanic	311 (22.2%)	1059 (19.4%)	0.019	32 (17.9%)	68 (18.6%)	0.842
Income < 50k	464 (33.2%)	1413 (25.9%)	<0.001	57 (31.8%)	106 (29.0%)	0.493
Parents: graduate degree	443 (31.7%)	2129 (39.1%)	<0.001	52 (29.1%)	118 (32.2%)	0.417

Categorical rows show one level; p is the omnibus test.

- Demographic selection occurs within the non-ADHD group (younger, male, Black, Hispanic, lower-SES children excluded); within the ADHD group, no demographic variable differs ($p \geq 0.10$).

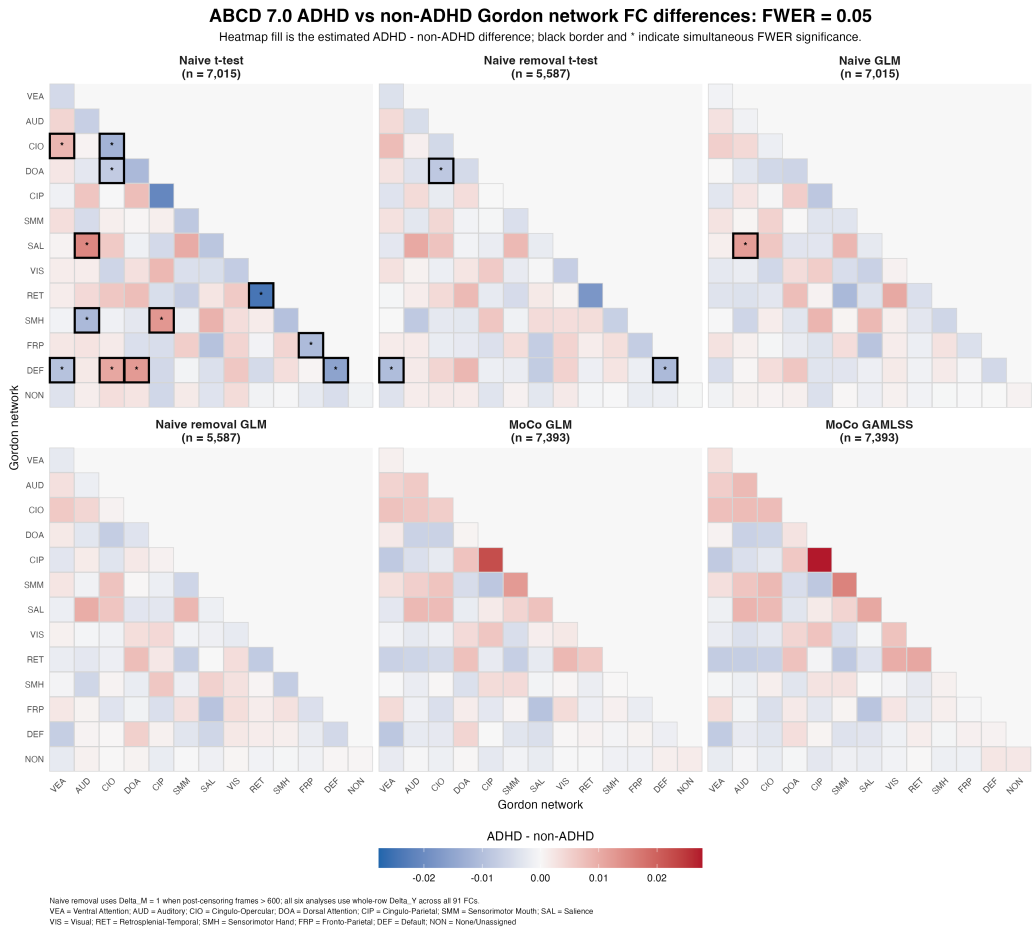
Selection bias in ABCD: symptoms and cognition (Z)

	Non-ADHD (n = 6,848)			ADHD (n = 545)		
Symptom / cognition (Z)	Excluded	Included	p	Excluded	Included	p
Crystallized cognition	50.46 (11.03)	52.15 (11.56)	<0.001	49.35 (11.23)	50.27 (11.37)	0.373
Fluid cognition	44.51 (11.33)	46.90 (11.01)	<0.001	40.88 (11.54)	43.62 (11.32)	0.009
KSADS inattention	0.56 (1.74)	0.32 (1.31)	<0.001	7.26 (1.71)	7.37 (1.64)	0.501
KSADS hyper/impulsive	0.34 (1.15)	0.18 (0.89)	<0.001	5.75 (2.38)	5.30 (2.48)	0.039
CBCL attention problems	53.77 (5.18)	52.81 (4.65)	<0.001	66.64 (8.84)	65.52 (8.80)	0.168
Sleep problems	36.88 (7.61)	35.89 (7.19)	<0.001	44.68 (12.32)	44.40 (11.52)	0.799
Mean FD	0.68 (0.50)	0.18 (0.15)	<0.001	0.86 (0.60)	0.20 (0.15)	<0.001

All rows: mean (SD). KSADS: present symptom counts.

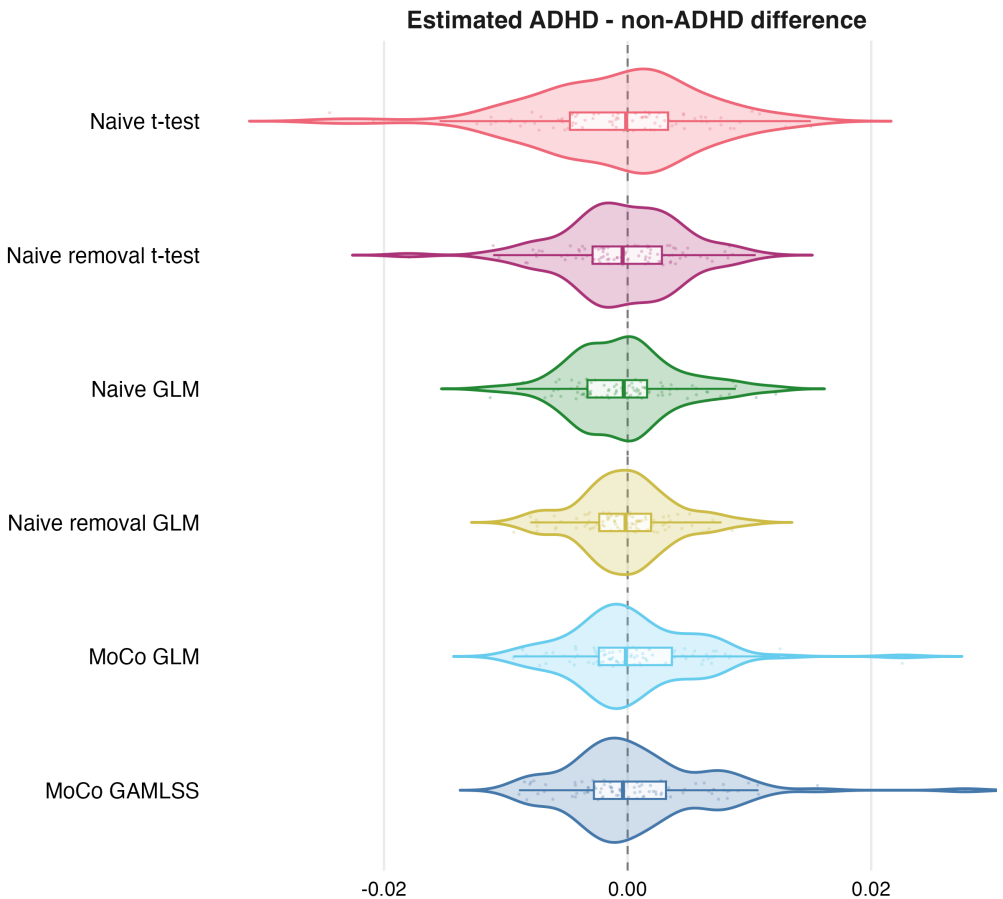
- ADHD children are excluded more often (32.8% vs. 20.4%), but within the ADHD group, excluded ≈ included on symptoms — exclusion does not select on ADHD severity here.
- Within the non-ADHD group, exclusion removes children with more subclinical symptoms and lower cognition, shifting the comparison-group composition.

ABCD results: network-pair differences (FWER = 0.05)



MoCo uses SuperLearner for the nuisance regressions. In the density estimations, MoCo (GLM density) uses a Gaussian density with log link, while MoCo (GAMLSS density) uses a gamma distribution with log link and includes splines for both the location and scale parameters.

ABCD results: ADHD vs. non-ADHD functional connectivity



Significant FCs among 91 Gordon network pairs (FWER = 0.05):

Method	N	Sig. FCs
Naive t-test	7,015	12
Naive removal t-test	5,587	3
Naive GLM	7,015	1
Naive removal GLM	5,587	0
MoCo (GLM density)	7,393	0
MoCo (GAMLSS density)	7,393	0

MoCo variants differ only in the estimator of the tolerable motion density $p_{M|\Delta=1,A=0,X}$.

ABCD interpretation

- Motion differs by diagnosis: children with ADHD move more (mean FD 0.41 vs. 0.29) and are excluded more often (32.8% vs. 20.4%).
- The naive t-test finds 12 significant FCs; adjusting for demographics and mean FD ($A + X + M$) leaves 1; removing high-motion children leaves 0–3; MoCo finds 0, where MoCo retains all 7,393 children and standardizes motion.
- Motion QC removal reduces power (21% of children lost; 33% of the ADHD group) and makes a "super control" non-ADHD group.
- Coherent account: the naive differences reflect motion artifact and demographic confounding; once motion is handled, ADHD–non-ADHD FC differences at ages 9–10 are small.
- Future work: pilot analysis of selection bias in ABCD children taking stimulants found that ADHD symptoms score was 30% higher in the excluded children, indicating substantial selection bias towards children with higher attention.

Acknowledgments

AI Statement: Claude assisted in the generation of the slides and figures used in this talk. Some figures are based on code that Claude wrote.

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